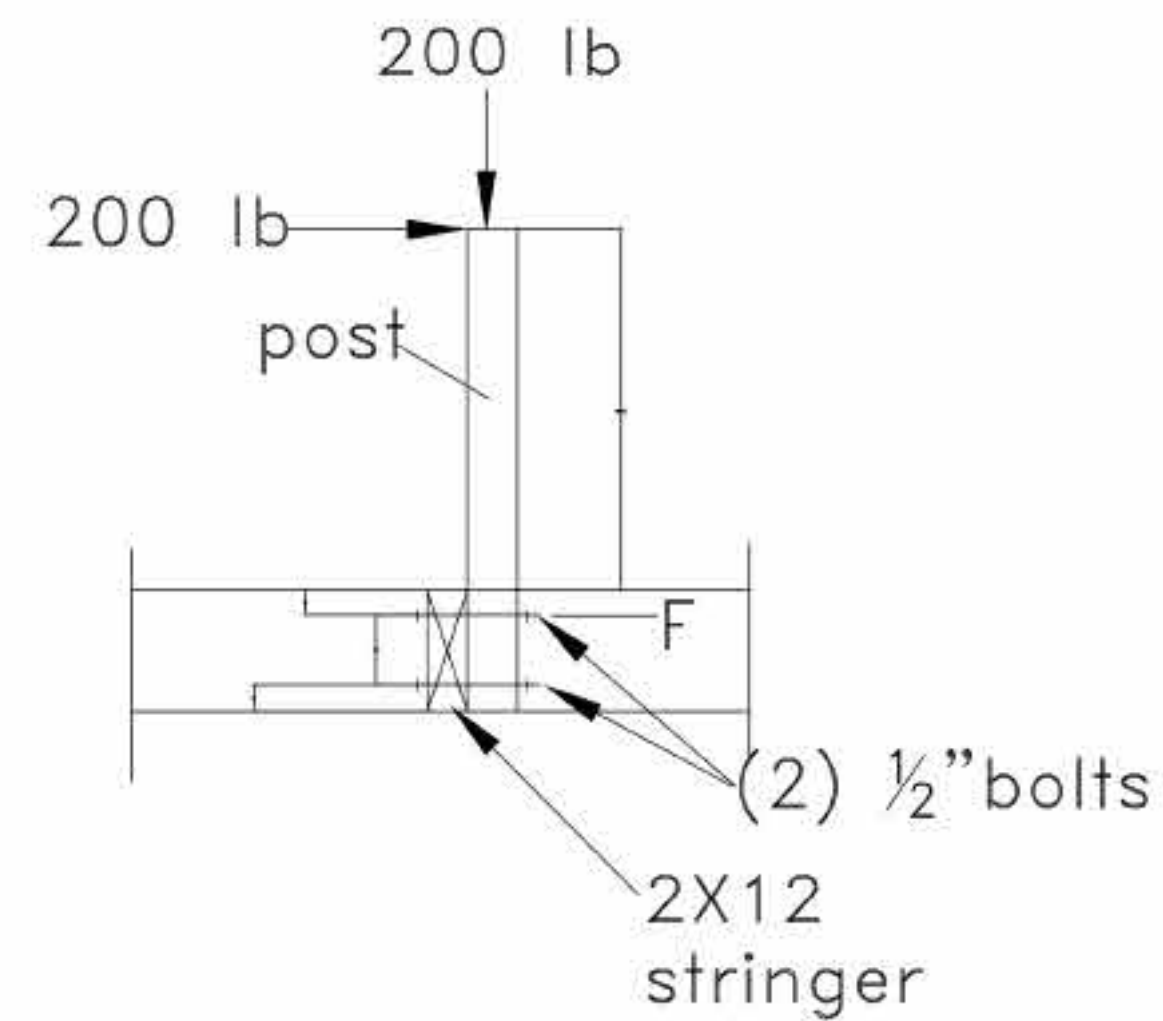




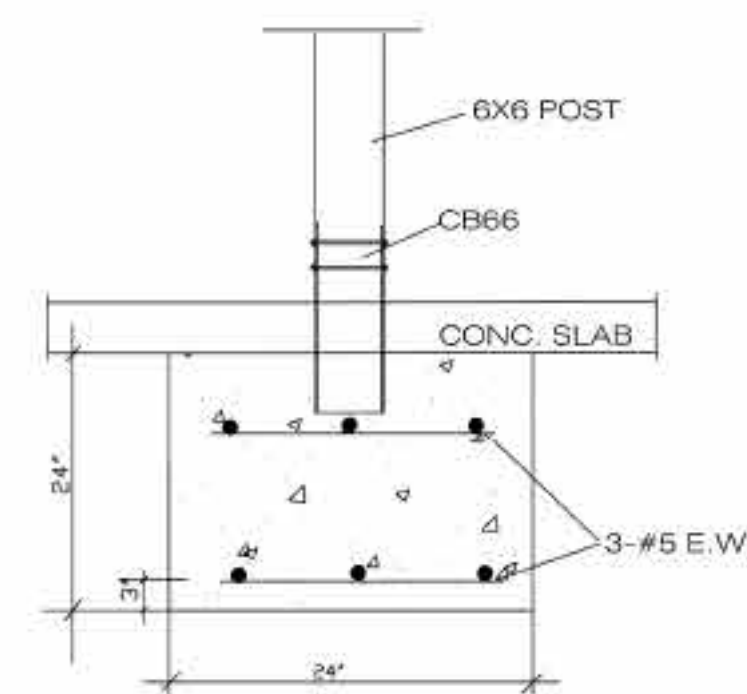
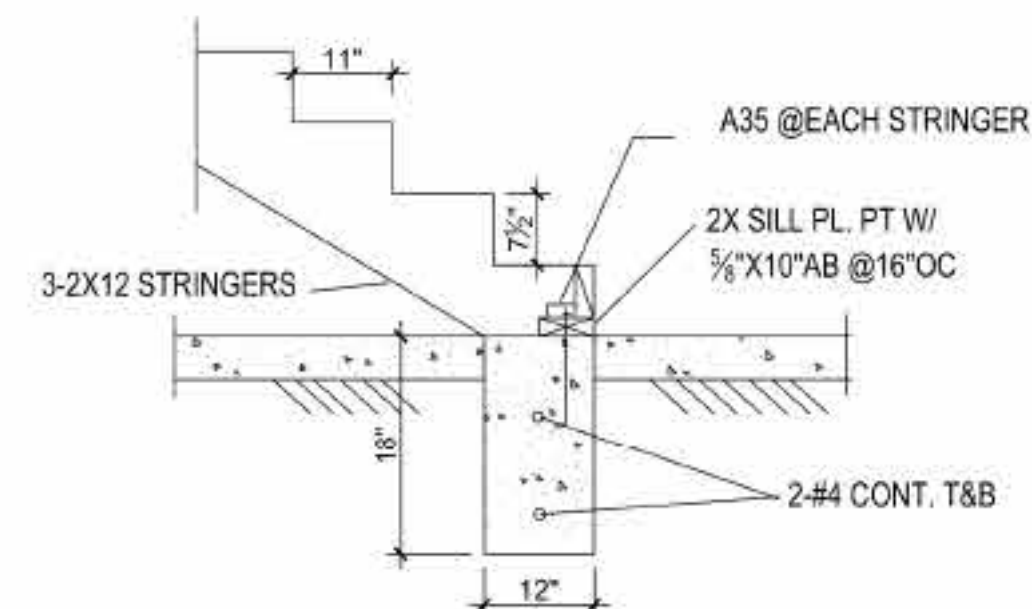
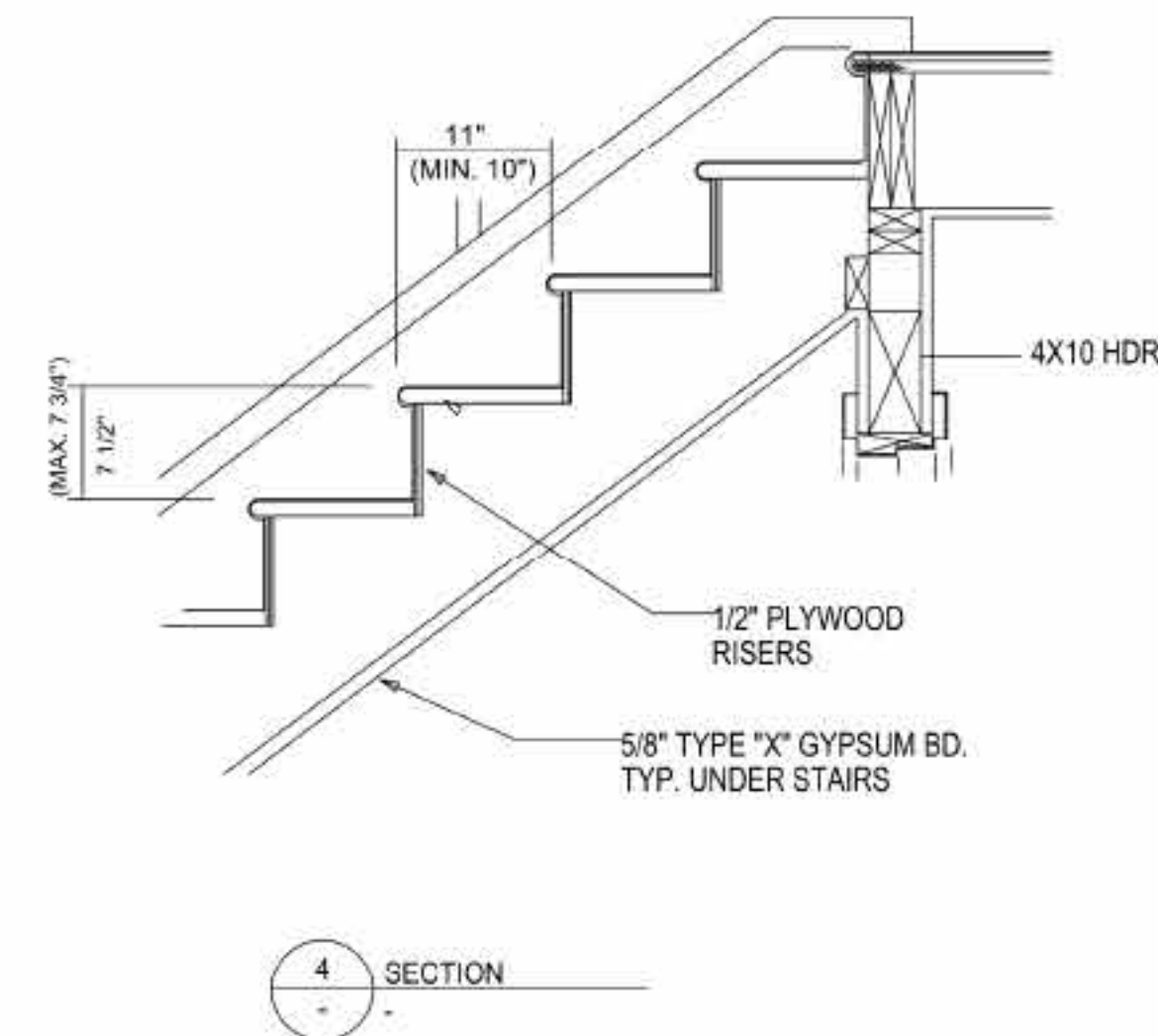
HANDRAIL CALS

P=200 lb (HORIZONTAL)
M = 200X48" = 9600 IN-LB
F = 9600 / 7.5" = 1280 LB
USE TWO 1/2" BOLTS A307
ALLOW Ft = 3900 lb
P=200 lb (VERTICAL AT POST)
ALLOW SHEAR = 550 lb (each bolt)

P=200 lb (VERTICAL AT CENTER SPAN)
w = 200/4 = 50 plf
M = (50X4²) / 8 = 100 ft-lb
R = 50X4/2 = 100 lb
2x4 A=5.25 S=3.06
fb = 100x12/3.06 = 392 psi
fv = 1.5x100/5.25 = 8.6 psi

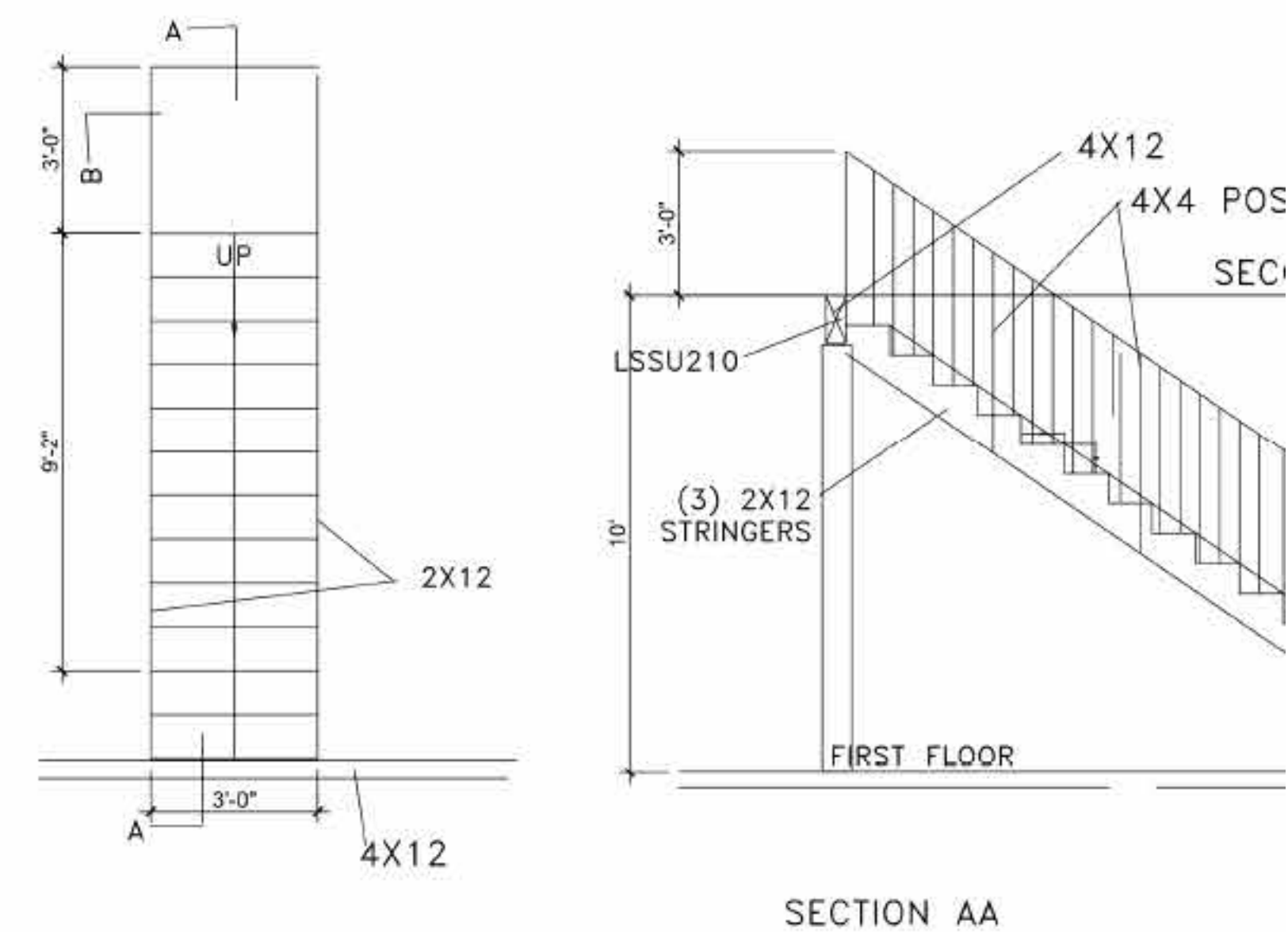
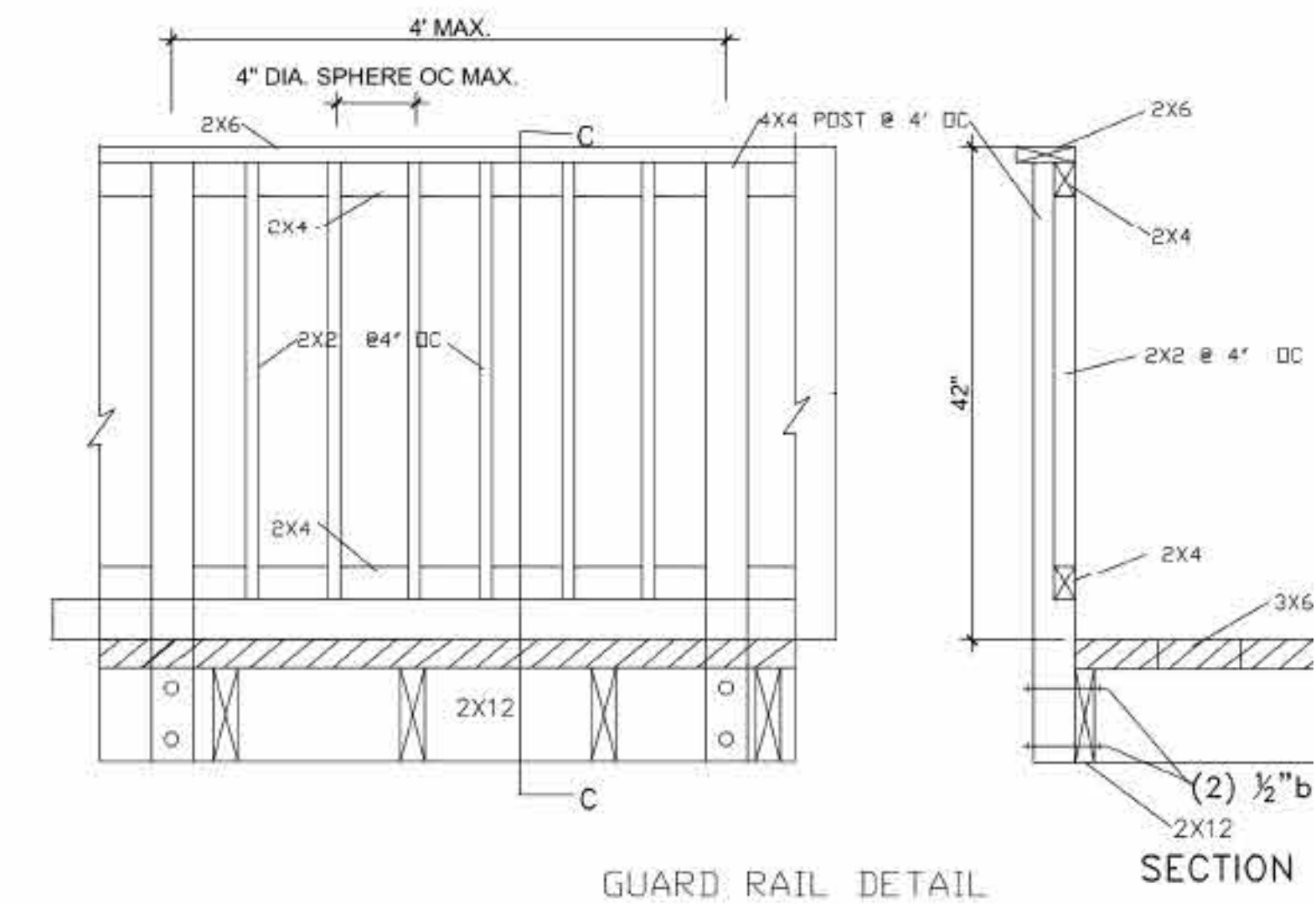
STAIR CALS

L=9'-2"
DL=12 PSFX11.1/9.2 = 14.5 PSF
LL = 40PSFX11.1/92 = 48.3 PSF
w = (14.5+48.3)x3.5/2 = 110 plf
M = 110X9.2²/8 = 1164 ft-lb
R = 110X9.2/2 = 506 LB
USE MINIMUM TWO 2X12 STRINGERS
S=31.64 A=16.88 I=178
DF-L NO. 2 Fb=900 psi Fv=180 psi E=1600 ksi
fb = 1164x12/31.64 = 441.5 psi
fv = 1.5x506/16.88 = 45 psi
DEFL = 5w⁴ / 384EI = 0.01"



FTG @ POST

2 COLUMN TO FTG



STAIR FRAMING PLAN

3 STAIR FRAMING DETAILS